

**Canada and World Issues:
A Geographic Analysis
Grade 12
University preparation
CGW4U**

MODULE: SUSTAINABLE DEVELOPMENT

Time: Minimum 5 Hours

Goal

Our main goal in producing this module is to develop a clear understanding of sustainable development, one that your students feel comfortable integrating into their own lives. Our objectives are divided into two categories. First, we will explore the behaviours and attitudes of your students towards adopting practices that are more sustainable. Second, we will broaden our scope to global initiatives in sustainable development by analysing the perspectives of various stakeholders in the sustainable development debate.

Goal:

To provide lesson plans using the Ontario Curriculum for Canadian and World Studies that will enable high school students to develop a clear understanding of sustainable development.

Objectives:

- To explore the behaviours and attitudes of your students towards adopting practices that are more sustainable.
- To analyse the perspectives of various stakeholders at the local, national and international level on sustainable development.

Module Description

In this module, students explore the challenges of creating a sustainable future through a critical analysis of sustainable development at the local, national and international level. Students will evaluate the approaches, policies and principles relating to environmental protection and the planet's life support systems placing specific emphasis on their own personal lives and how their actions impact the sustainability of the planet.

This module provides a firm foundation for understanding the terminology and key issues pertaining to sustainable development. Furthermore, a number of geographic methods are used to introduce students to the issues surrounding sustainable development. These include data analysis, issue analysis, and mapping techniques. Furthermore, the lesson plans incorporate a variety of teaching and learning methods to provide a stimulating environment for the learner to gather knowledge and encourage critical and analytical debate on the issue of sustainable development. The use of simulation, role play, group discussion and a number of other participatory techniques are used throughout the module.

The module is self-containing meaning that all the resources needed to use the lesson plans are included. Should you be interested in exploring the issue further, a number of references are provided for your convenience.

Defining Sustainable Development

The term sustainable development has been interpreted in many different ways. The most common definition comes from the Brundtland Report, published by the UN World Commission on Environment and Development in 1987.

Definition of Sustainable Development

“Development which meets the needs of present generations without compromising the ability of future generations to meet their own needs.” Our Common Future: Report of the World Commission on Environment and Development (The Brundtland Report), 1987

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An eight point outline of sustainable development was designed by the Sustainability Project, a non-profit initiative in Ontario aiming to:

- clarify the goal of sustainability through education, debate and public participation, and to
- promote sustainability as a goal we can choose.

The eight points are meant to be a guideline for sustainable development:

Eight Point Outline of Sustainable Development	
Sustainable Activities	Unsustainable Activities
• Use materials in continuous cycles (Re-using materials either through natural processes or recycling programs). • Use continuously reliable sources of energy. (Heat from the Earth's core, tides, the sun (nuclear fusion at a safe distance) and the wind and water which the sun sets in motion). • Come mainly from the qualities of being human (i.e. creativity, communication, coordination, appreciation, and spiritual and intellectual development).	• Require continual inputs of non-renewable resources (Resources available only in limited quantity. Metals, coal and oil are notable examples). • Use renewable resources faster than their rate of renewal (Renewable resources: grow and increase through natural processes; e.g. forests, fish stocks, ground water and soil fertility. The rate at which they are used $\leq$ the rate at which they grow or accumulate). • Cause cumulative degradation of the environment (Creating waste which nature cannot handle, or which cannot be absorbed as fast as we create it). • Require resources in quantities that undermine other people's well-being (People taking unfair advantage of others). • Lead to the extinction of other life forms (The loss of species means we also lose genetic possibilities for fighting disease, in people and in food crops, as well as potential new sources of food).

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¹ World Commission on Sustainable Development. 1987. Our Common Future. Oxford: Oxford University Press.

² The Sustainability Project. 1999. www.cyberus.ca/~sustain1/Sustain.html

The Sustainable Development Timeline

In 1962, *Silent Spring* was published, a book many consider a turning point in our understanding of the interconnections between the environment, economy and social well-being. In the decades that have followed, many milestones have marked the journey toward sustainable development.

The United Nations Conference on the Human Environment, which was held in Stockholm in 1972 and attended by 113 nations, represented the beginning of concerted global efforts to deal with environmental problems. At the Conference, the Declaration on the Human Environment was adopted to "inspire and guide the peoples of the world in the preservation and enhancement of the human environment." The proclamations and principles contained within this declaration addressed the issues of human impact on the environment, the role of governments, population growth, underdevelopment, assistance to developing nations to meet environmental objectives, the necessity of continued social and economic development, the role for international cooperation and organisations, and the importance of science, technology, and education. The link was made between quality of life and quality of environment for present and future generations.³

Following the UN Conference on Human Development, a number of countries began to form national environmental protection agencies. In 1987, "Our Common Future" (the Brundtland Report) was published by the World Commission on Environment and Development. It was in this report that the term "sustainable development" was popularised.

Below is a chart that summarises some of the significant events in the journey toward sustainable development. More information can be found in Appendix 1 and on the Sustainable Development Gateway Website (<http://sdgateway.net/introsd/timeline.htm>).

THE TIMELINE FOR SUSTAINABLE DEVELOPMENT

THE UN CONFERENCE ON HUMAN DEVELOPMENT

What: UN Conference on Human Development

Where: Stockholm

When: 1972

Detail: The first time the international community met to consider the global environment and development needs

Result: The Stockholm Declaration and Action Plan was developed, which defined the principles for the preservation and enhancement of the natural environment and highlighted the need to support people in this process.

Source: <http://www.unep.org/>

OUR COMMON FUTURE

What: The Brundtland Commission or the UN World Commission on the Environment and Development

When: 1983

Detail: The Commission was set up to focus on the causes of environmental problems rather than the effects of environmental degradation.

Result: Framed much of what would become the 40 chapters of Agenda 21 and the 27 principles of the Rio Declaration on Environment and Development.

Source: World Commission on Sustainable Development. 1987. *Our Common Future*. Oxford: Oxford University Press.

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³ Environment Canada. Sustainable Development Information. http://www.sdinfo.gc.ca/historical_path/index_e.cfm?id=106

CANADA'S RESPONSE TO OUR COMMON FUTURE

What: The creation of the National Round Table on the Environment and the Economy

Where: Canada

When: 1989

Detail: Comprised of representatives from government, industry, science, education, NGOs and native peoples to advise the Prime Minister of Canada on policies for Sustainable Development and to develop strategies for action. Learning for a Sustainable Future, a sustainable development education program is a spin off of the round table (<http://www.schoolnet.ca/learning/>).

Source: <http://www.nrtee-trnee.ca>

THE EARTH SUMMIT

What: The UN Conference on Environment and Development or the Earth Summit

Where: Rio de Janeiro

When: 1992

Detail: Accepted Agenda 21 and the Rio Declaration, which outlined key policies for achieving sustainable development that meets the needs of the poor and recognizes the limits of development to meet global needs.

Result: Brought the environment and development into the public arena and led to the development of two legally binding documents; a) Biological diversity and b) United Nations Framework Convention on Climate Change. Led to the formation of the Commission for Sustainable Development (CSD).

Source: <http://www.un.org/geninfo/bp/enviro.html>

AGENDA 21

What: A comprehensive plan of action to be taken globally, nationally and locally by organizations of the United Nations System, governments, and major groups in every area in which humans impact on the environment.

Where: Rio de Janeiro, Brazil

When: 1992

Detail: Outlines strategies and integrated program measures to halt and reverse the effects of environmental degradation. It promotes environmental renewal and sustainable development in all countries.

Agenda 21, the Rio Declaration on Environment and Development, and the Statement of principles for the Sustainable Management of Forests were adopted by more than 178 governments at the United Nations Conference on Environment and Development (UNCED).

Agenda 21 called for a total shift in the status quo of value systems and institutional processes.

Source: <http://www.un.org/esa/sustdev/agenda21.htm>

THE INTERNATIONAL INSTITUTE FOR SUSTAINABLE DEVELOPMENT

What: IISD is an organization that promotes the transition toward a sustainable future. It seeks to demonstrate how human ingenuity can be applied to improve the well-being of the environment, economy and society.

When: 1988

Where: Winnipeg, Manitoba

Detail: In 1988, Prime Minister Brian Mulroney announced, to the General Assembly of the United Nations the creation of the International Institute for Sustainable Development in Winnipeg. IISD officially opened in March 1990.

Source: www.iisd.ca/common/host.htm

Rio + 5

What: The Earth Summit Review

Where: New York

When: 1997

Detail: The goal of Rio + 5 was to assess the progress made since the 1992 Earth Summit. Several gaps were identified with regard to social equity and poverty due to falling official development assistance to poor countries and growing international debt.

Result: Called for stronger implementation of the growing number of international agreements and conventions which pertain to the environment and development.

Source: <http://www.un.org/esa/earthsummit/>

THE EARTH SUMMIT 2002

What: Rio' Earth Summit +10

Where: Johannesburg, South Africa

When: August 26, 2002

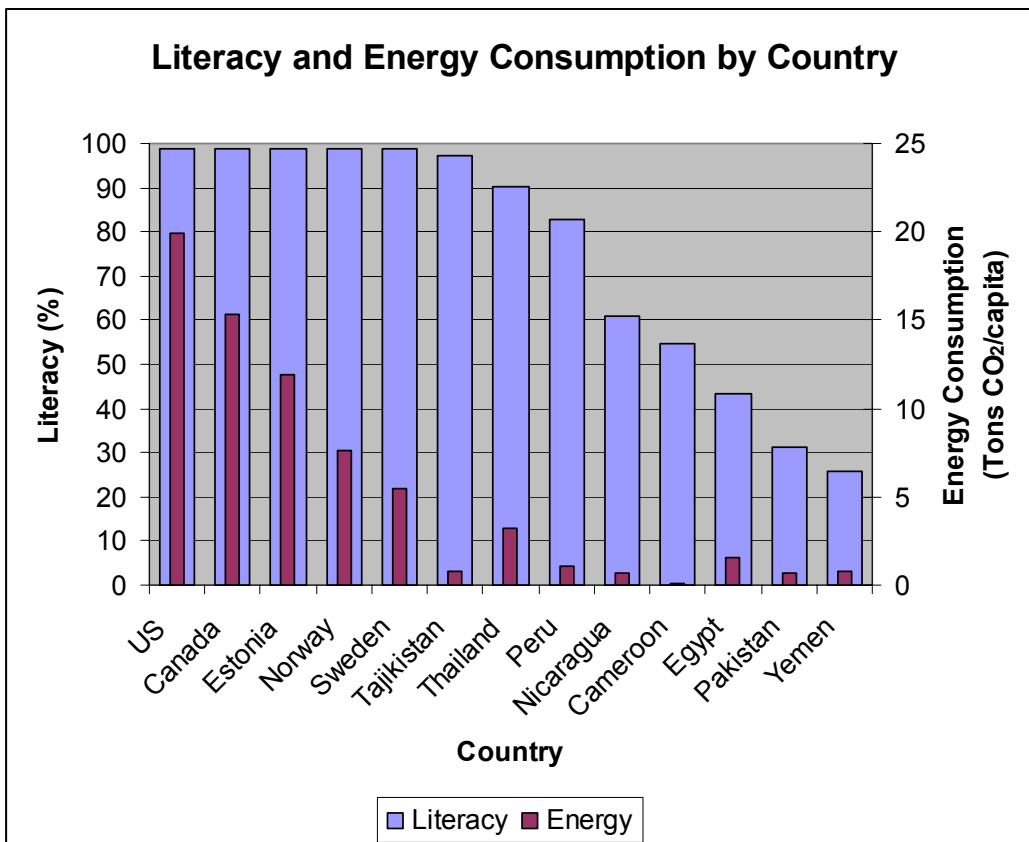
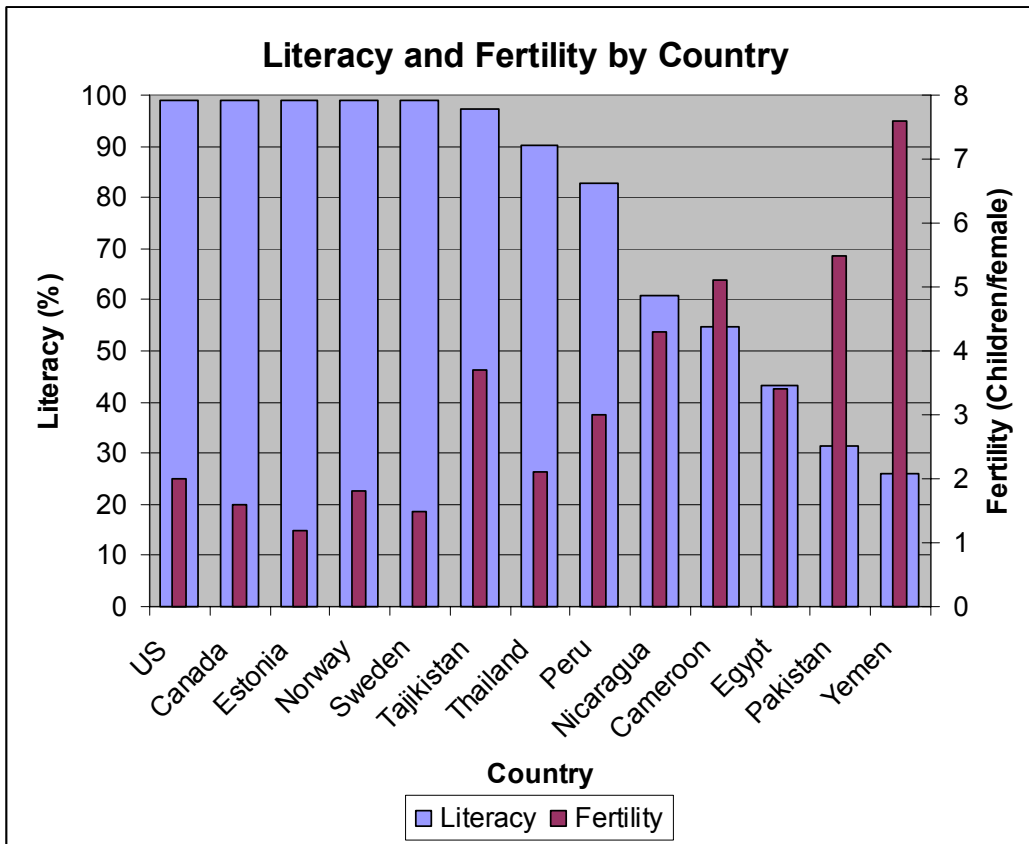
Detail: The goal of the summit was to identify concrete steps for moving forward on a number of priority areas of sustainable development.

Anticipated Results: The Summit reaffirmed sustainable development as a central element of the international agenda and gave new impetus to global action to fight poverty and protect the environment. Governments agreed to and reaffirmed a wide range of concrete commitments and targets for action to achieve more effective implementation of sustainable development objectives.

Source: <http://www.johannesburgsummit.org/index.html>

The achievement of sustainable development is dependent on two major components of human activity – population and resource consumption. An increase in population and resource consumption is considered to be a threat to sustainable development. The education level of individuals affects both fertility rates and resource consumption. As women become more educated, statistics show that fertility rates decline, thereby decreasing the threat of overpopulation. This in turn facilitates the progress toward sustainable development. Research has shown that women with more education tend to have more status in the community. They are also more likely to marry later and have greater bargaining power in the household. All of these factors seem to result in women having smaller families. (See graph comparing fertility rates to education level). Women have traditionally had large families in order to secure their livelihood in their old age. Women with more education recognize that their livelihood is still secure when their families are small. Instead of equating livelihood with more children, these women associate livelihood with having greater career and educational objectives for their children. In this case, more education facilitates progress toward sustainability.

In some cases, more education can *increase* the threat to sustainability. Higher education usually coincides with higher incomes allowing for a greater disposable income with which to consume resources. Statistics show that education and per-capita energy use and waste generation are correlated (See graph comparing energy consumption to education level). This suggests that education alone will not lead to greater sustainability. Rather, there is a need to provide education on resource use and consumption patterns.



⁴ United Nations Development Programme. 2002. Human Development Report 2002. UNDP, New York.

Sustainable Development in the Classroom

It is with this in mind that we have designed this module. Your students most likely live with an abundance of natural resources. The thought of shortages seems inconceivable, yet many of our seemingly limitless resources are depleting. For example, most freshwater sources in Canada have diminished substantially in recent years due to drought and warm temperatures. There is an immediate need to change how we use water and other resources in Canada if we are to make them sustainable. This change in behaviour will require education.

Education was identified in the first Earth Summit in 1992 as having primary responsibility for implementing the changes necessary for sustainable development: "Education is the single most important influence in changing human behaviour, promoting economic growth and increasing quality of life, providing the knowledge and skills that produce jobs and increase productivity. It equips people for meeting needs."⁵

To meet this objective, local schools can incorporate sustainable development into existing Curriculum. Sustainable development is a concept that is recognised in the Ontario Curriculum for Canadian and World Studies (2000), specifically in the Grade 12 Canada and World Issues Course (See Synopsis Chart on the next page). While the units that follow emphasise links with the Canada and World Issues Course of the Ontario Curriculum, the material is applicable to other courses and Curriculum guides across the country.

It should be noted that a single approach to sustainable development does not exist. Rather, the approach taken by you and your students should be relevant both locally and culturally to your classroom environment. We have provided some activities for you and your students to develop your own classroom charter for sustainable development (See Unit 1: Activity 3)

⁵ Learning for a Sustainable Future. 1999.
<http://www.schoolnet.ca/future/teacher/classroom/manitoba/sustdev/education/content.htm>